Sensors and State Estimation

Course notes – Autumn 2022/23

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Source files:

- `22_23_UAVs_Ch5-6_Sensors_and_State_Estimation_1.pdf` (28 pages) - course lecture, Técnico Lisboa, MEAer UAVs, Autumn 2022/2023, by Rita Cunha / J.R. Azinheira
- `22_23_UAVs_Ch5_Sensors_Beard_McLain.pdf` (29 pages) - reference chapter

(Ch. 7 of the source textbook)

- `22_23_UAVs_Ch6_State_Estimation_Beard_McLain.pdf` (47 pages) - reference chapter (Ch. 8 of the source textbook)

Both reference PDFs are slide adaptations of Randal Beard & Timothy McLain, *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, 2012 (Ch. 7 "Sensors" and Ch. 8 "State Estimation"). All pages processed (28/28, 29/29, 47/47).

1 Course lecture (Ch5-6 Sensors and State Estimation)

The lecture is built directly on top of the Beard & McLain slide deck and adds annotations specific to the AR.Drone platform used in the course labs, plus a worked observability example and a treatment of complementary filters.

1.1 AR.Drone-specific sensors (course addition, not in the textbook)

- The AR.Drone has **no GPS**. Its extra sensors compared to the generic textbook list are:
  - **Sonar** (ultrasonic): 40 kHz resonance, range up to 6 m, sampled at 25 Hz. Used to estimate height/vertical displacement and depth of the scene seen by the vertical (downward-facing) camera.
  - **Vision** (downward camera, hovering flight): speed estimation via two alternative algorithms:
    1. Multi-resolution optical-flow scheme over the whole image.
    2. "Corner tracking" - track displacement of a set of interest points (trackers).

1.2 Accelerometers - basic principle (spring-mass-damper derivation)

Derived from a proof mass on a spring/damper inside a casing:

- h: inertial height of casing; $h1 = h + \delta$: inertial height of proof mass; δ: relative displacement (what is actually measured).
- Newton's 2nd law on the proof mass: $m \cdot \ddot{h}1 = -k(h1 - h) - \beta(\dot{h}1 - \dot{h}) - mg$.
- Substituting $z = -h$, this becomes the accelerometer 2nd-order dynamics: $m \cdot \ddot{\delta} + \beta \cdot \dot{\delta} + k \cdot \delta = m(\ddot{z} - g)$.
- Laplace/transfer function: $D(s) = 1/(s^2 + (\beta/m)s + k/m) \cdot (Az(s) - g/s)$.
- Low-frequency approximation: $\delta \approx (m/k)(\ddot{z} - g)$ - i.e. below the sensor's natural frequency, deflection is proportional to (acceleration minus gravity).
- For a 3-axis accelerometer rigidly attached to the vehicle body frame {B}: $\delta \propto R^T(\ddot{p} - g \cdot e3) = S(\omega)v + \dot{v} - g \cdot R^T e3$ (S(ω) is the skew-symmetric matrix of the angular velocity - same notation as the rigid-body chapter).

- Near hover (small p̈, small ω): $\delta \propto -g \cdot R^T e3 = -g \cdot [-sin\theta, cos\theta \cdot sin\varphi, cos\theta \cdot cos\varphi]^T$.
- **Key takeaway**: accelerometers alone give roll/pitch information only near hover / unaccelerated flight - they are ”typically used for attitude estimation in combination with other sensors” (fused with gyros/GPS).

1.3 Rate gyros - basic principle (Coriolis / MEMS vibrating mass)

- A vibrating proof mass with velocity v on a rotating body experiences a Coriolis acceleration aC. For a package rotating with $\omega = [0, 0, \Omega]^T$ and vibration velocity $v = [vx, 0, 0]^T$, $\ddot{p}1 = -[0, 2\Omega \cdot vx, 0]^T + R^T \ddot{p}$ - the second term is proportional to the angular rate Ω being measured. (Detailed kinematic derivation: $p1 = R^T p$, $\dot{p}1 = -S(\omega)R^T p + R^T \dot{p}$, $\ddot{p}1 = -S(\dot{\omega})R^T p + S(\omega)^2 R^T p - 2S(\omega)R^T \dot{p} + R^T \ddot{p}$.)

1.4 Magnetometers - heading recovery

- $Im0 = [cos\delta \cdot cos\gamma, sin\delta \cdot cos\gamma, sin\gamma]^T$ is the known magnetic field vector in the inertial frame (δ = declination, γ = inclination, location-dependent).
- Body-frame reading: $Bm0 = BR_I \cdot Im0$, and roll/pitch (φ,θ) are assumed known from other sensors.
- Compensating for roll and pitch: $B1m0 = Ry(\theta)Rx(\varphi) \cdot Bm0 = Rz(-\psi) \cdot Im0$, which reduces to a 2D rotation by $(\psi - \delta)$ in the horizontal plane.
- Closed form for heading: $(\psi - \delta) = \text{atan2}(-B1my, B1mx)$.

1.5 Kalman filter (course treatment)

- Continuous-time stochastic LTI model: $\dot{x} = Ax + Bu + w$, $y = Cy + n$ (should read $y = Cx + n$), with w, n zero-mean white noise, $E[w(t)w(\tau)^T] = \delta(t-\tau)W$, $E[n(t)n(\tau)^T] = \delta(t-\tau)N$.
- Objective: minimize $J = lim_{t\to\infty} E[\tilde{x}(t)^T \tilde{x}(t)] = limtr[\Sigma(t)]$, with error covariance $\Sigma(t) = E[\tilde{x}(t)\tilde{x}(t)^T]$.
- Observer form: $\dot{\hat{x}} = A\hat{x} + Bu - L(y - C\hat{x})$, gain $L = \Sigma\infty \cdot C^T \cdot N^{-1}$, where $\Sigma\infty$ solves the steady-state Riccati-like equation $(A - LC)\Sigma\infty + \Sigma\infty(A^T - C^T L^T) + W + LNL^T = 0$.
- **Duality LQR ↔ Kalman filter** made explicit side by side:
  - LQR: $\dot{x} = Ax + Bu$, $u = -Kx$, cost $J = \int (x^T Qx + u^T Ru)dt$, $Q = C^T C \geq 0$, $R > 0$, gain $K = R^{-1}B^T P$, $PA + A^T P + Q - PBR^{-1}B^T P = 0$. Requires (A,B) stabilizable, (A,C) detectable.
  - KF: $\dot{x} = Ax + Bu + \bar{B}w$, $y = Cx + n$, gain $L = \Sigma C^T N^{-1}$, $\Sigma A^T + A\Sigma + W - \Sigma C^T N^{-1} C\Sigma = 0$. Requires (A,C) detectable, $(A, \bar{B})$ controllable (marked ”?” as an open point in slides, i.e. worth double-checking case by case).

1.6 Worked example: roll-angle estimation for a fixed-wing aircraft

Assumptions: coordinated turn, no wind, no sideslip. From force balance: $Flift \cdot cos\varphi = mg$, $Flift \cdot sin\varphi = mV^2/R = mV \cdot \dot{\psi}$, giving $tan\varphi = V\dot{\psi}/g$, i.e. $\dot{\psi} =$

$(g/V) tan\varphi$.

The course walks through **three progressively richer Kalman filter setups** for this example, illustrating the role of observability:

1. **Rate-gyro only, state = [φ, bp, br]**, output $y = rm$ (yaw-rate measurement only): model $\dot{\varphi} = pm - bp - np$, $\dot{bp} = n1$, $\dot{br} = n2$, output $y = rm = (g/V)\varphi + br + nr$. → **Not observable** (pair (A,C) fails the observability test) - additional sensor is needed. This is a nice didactic "what goes wrong" case.
2. **Rate gyros + accelerometer, state = [φ, p, bp, br]**, outputs $y = [rm, pm, \varphi m]^T$ (adds roll-rate measurement `pm` and a (noisy) direct roll measurement φm from the accelerometer, derived from $am \approx -g[0, sin\varphi, cos\varphi]^T$ under the coordinated-turn assumption, i.e. $am = -g[0, 0, \sqrt{(}1 + tan^2\varphi)]$ after substitution). → **Observable** - this is the version that actually works.
3. A **second alternative** with the same state/output structure but using the dynamic model for the time evolution of $\dot{\varphi}$ directly (equivalent result, different parametrization).

1.7 Complementary filters

- **Context**: different sensors are reliable over different (complementary) frequency bands - e.g. GPS velocity is accurate at low frequency but slow/noisy at high frequency, while integrated accelerometer data is accurate at high frequency but drifts (bias accumulates) at low frequency.
- **Structure**: $\hat{x} = GL(s) \cdot xmL + GH(s) \cdot xmH$, with $GL(s) = l/(s + l)$ (low-pass) and $GH(s) = s/(s + l)$ (high-pass), satisfying $GL(s) + GH(s) = 1$ for all s.
- **Worked example - altitude-rate estimation**: $\dot{h}GPS = \dot{h}(t) + n1(t)$ (low-frequency-good) and $\ddot{h}ACC$ integrated once to $\dot{h}ACC = \dot{h}(t) + n2(t)$ (high-frequency-good). Combined estimate: $\hat{H}(s) = l/(s + l) \cdot \dot{H}1(s) + s/(s + l) \cdot \dot{H}2(s) = \dot{H}(s) + l/(s + l) \cdot N1(s) + s/(s + l) \cdot N2(s)$ → keeps the whole true signal, attenuates N1 at high frequency and N2 at low frequency.
- **Connection to Kalman filter**: shown to be an exact special case. With $x = \dot{h}$, $u = \ddot{h}ACC$, $y = \dot{h}GPS$, the KF observer $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$ reduces to exactly the complementary filter structure - i.e. the complementary filter is a Kalman filter with fixed (non-adaptive) gain `l`.
- **More advanced case shown**: a state observer for **roll angle + roll-rate bias**, $x = [\varphi, bp]$, $u = pm$, $y = \varphi m$, producing a second-order complementary filter $\hat{\Phi}(s) = GL(s)\Phi m(s) + GH(s)Pm(s)/s$ with $GL(s) = (l1s + l2)/(s^2 + l1s + l2)$.

2 Textbook reference - Sensors (Beard & McLain Ch5, "Chapter 7" in the original book numbering)

Covers, in order: accelerometers, rate gyros, pressure sensors (absolute + differential), magnetometers/compasses, GPS.

2.1 Accelerometers (MEMS)

- Physical model: proof mass m on spring k, $m\ddot{x}+kx = k(y-x)$ where y is casing displacement, x proof-mass deflection. Laplace: $X(s)/Y(s) = 1/((m/k)s^2+1)$, equivalently in terms of acceleration $AX(s)/AY(s) = 1/((m/k)s^2+1)$.
- Model with bias + noise: $\Upsilon accel = kaccel \cdot a + \beta accel + \eta' accel$.
- **Key conceptual point (repeated, "tricky concept")**: an accelerometer measures **specific force**, i.e. $a = (1/m)(Ftotal - Fgravity)$, or equivalently $ameasured = (1/m)(\Sigma Fnon - gravitational)$. **Accelerometers do not measure gravity** - a stationary accelerometer on a table reads +g upward (reaction force), not zero, because the proof mass and casing are NOT both free-falling together in that scenario (unlike free fall, where both experience gravity identically and the reading is zero).
- For a fixed-wing aircraft: $ameasured = (1/m)(Flift + Fdrag + Fthrust)$ (gravity cancels analytically).
- Full body-frame accelerometer equations (from Ch.3 rigid-body dynamics $m(dv/dt_b + \omega_{b/i} \times v) = Ftotal$):
  - $ax = \dot{u} + qw - rv + g \cdot sin\theta$
  - $ay = \dot{v} + ru - pw - g \cdot cos\theta \cdot sin\varphi$
  - $az = \dot{w} + pv - qu - g \cdot cos\theta \cdot cos\varphi$ With sensor noise added: $y_accel, i = (kinematicterm) + \eta_accel, i$.
- Also expressed in terms of aerodynamic coefficients (CX, CY, CZ, angle of attack α, sideslip β, control surfaces $\delta a, \delta e, \delta r$, motor throttle δt) - the full nonlinear aerodynamic force model, useful for high-fidelity simulation.
- Compact "specific force" form: $y_accel, x = fx/m + g \cdot sin\theta + \eta$, etc. (fx,fy,fz = non-gravitational body forces).

2.2 MEMS rate gyro

- Physical principle: a point translating on a rotating rigid body experiences Coriolis acceleration $aC = 2\Omega \times v$. A resonating proof mass with $|v| = A \cdot \omega n \cdot \sin(\omega n \cdot t)$ yields a sensor output $Vgyro = kC|aC| = 2kC \cdot \Omega \cdot |A\omega n \cdot \sin(\omega nt)| \to 2kC \cdot A \cdot \omega n \cdot \Omega = KC \cdot \Omega$ (proportional to angular rate).
- Model: $\Upsilon gyro = kgyro \cdot \Omega + \beta gyro(bias, frommanufacturing/drift) + \eta' gyro$ (zero-mean Gaussian noise). Calibrated: $y_gyro, x = p + \beta gyro, x + \eta gyro, x$ (and similarly for q,r).

2.3 Pressure sensors

- Piezoresistive diaphragm: external pressure deflects a thin diaphragm, measured via a doped piezoresistor bridge.
- **Absolute pressure → altitude**: hydrostatics $P2 - P1 = \rho g(z2 - z1)$. Using ground as reference: $P - Pground = -\rho g \cdot h_AGL$. Below 11 km, barometric formula with temperature lapse rate accounts for density-with-altitude: $P = P0 \cdot [T0/(T0 + L0 \cdot h_ASL)]^(gM/RL0)$, with $L0 = -0.0065K/m$. Simplified constant-density sensor model: $y_abs_pres = \rho g \cdot h_AGL + \beta abs + \eta abs$. (Slides

show that the constant-density approximation diverges badly from the ideal-gas-law barometric formula above 2000 m, but is fine below 500 m.)

- **Differential (dynamic) pressure → airspeed**: Bernoulli, $Pt = Ps + \rho Va^2/2 \Rightarrow \rho Va^2/2 = Pt - Ps$. Sensor model: $y_diff_pres = \rho Va^2/2 + \beta diff + \eta diff$. Implemented via a pitot-static tube (measures total pressure Pt at the tube tip and static pressure Ps from side ports, feeding a differential-pressure diaphragm sensor).

2.4 Magnetometers / digital compasses

- Heading = declination + magnetic heading: $\psi = \delta + \psi m$.
- $m0v1 = Rbv1(\varphi, \theta) \cdot m0b = Rv2v1(\theta) \cdot Rbv2(\varphi) \cdot m0b$, explicit rotation-matrix expansion given in slides using (cθ,sθ,cφ,sφ) shorthand.
- Closed form: $\psi m = -\text{atan2}(m0y^v1, m0x^v1)$.
- Magnetic **inclination** (dip angle): angle of the field vector below horizontal; positive = pointing into the Earth. Both declination and inclination vary geographically (world maps shown) - relevant for accurate heading estimation depending on operating location.

2.5 GPS

- Constellation: 24 satellites, altitude 20,180 km; any point on Earth sees ≥4 at all times. 4 pseudorange measurements needed (3 position unknowns + 1 receiver clock-offset unknown) → 4 nonlinear equations in 4 unknowns (lat, lon, alt, clock offset).
- **Error sources**: ephemeris data, satellite clock drift, ionospheric delay, tropospheric delay, multipath, receiver measurement noise. Rule of thumb: 10 ns timing error → 3 m pseudorange error.
- **UERE** (User Equivalent Range Error) combines bias + random components per source (table given: ephemeris 2.1/0.0, satellite clock 2.0/0.7, ionosphere 4.0/0.5, troposphere 0.5/0.5, multipath 1.0/1.0, receiver 0.5/0.2 → total UERE rms ≈ 5.3 m, filtered ≈ 5.1 m).
- **DOP** (Dilution of Precision): satellite geometry effect on position accuracy; close-together satellites → high DOP (bad), spread-out → low DOP (good). Nominal HDOP=1.3, VDOP=1.8.
- Total error: $E_n - e, rms = HDOP \times UERE_rms$ (≈6.6 m), $E_h, rms = VDOP \times UERE_rms$ (≈9.2 m).
- **Gauss-Markov error model** (Rankin) for GPS error time-evolution: $\nu[n+1] = e^(-kGPS \cdot Ts) \cdot \nu[n] + \eta GPS[n]$, first-order correlated random walk. Parameters given per axis (North/East/Altitude): nominal bias/random 1-σ errors, ηGPS std dev, correlation time $1/kGPS = 1100s$, $Ts = 1.0s$. Measurement model: $yGPS, n[n] = pn[n] + \nu n[n]$ (similarly for east, altitude with sign flip since altitude = -pd).

3 Textbook reference - State Estimation (Beard & McLain Ch6, "Chapter 8" in the original book numbering)

3.1 Motivation and architecture

- Standard UAV autopilot architecture chain: **Path planner → Path manager → Path following → Autopilot → Unmanned Vehicle**, all fed by a **State estimator** block that consumes on-board sensor data and outputs $\hat{x}(t)$ to every other block.
- Not all states are directly measured. Table of what's measured vs. estimated:
  - `pn, pe`: measured (GPS) and smoothed.
  - `pd` (altitude): measured (absolute pressure, GPS).
  - `u, v, w` (body velocities): estimated with EKF.
  - φ, θ, ψ (attitude): estimated with EKF.
  - `p, q, r` (angular rates): measured directly (rate gyro).

3.2 Sensor-model inversion (cheap "quasi-estimation" before resorting to filtering)

- Angular rates: $\hat{p} = LPF(y_gyro, x)$ etc. (just low-pass filter the raw gyro, correcting scaling).
- Altitude: $\hat{h} = LPF(y_static_pres)/(\rho g)$.
- Airspeed: $\hat{V}a = \sqrt((2/\rho) \cdot LPF(y_diff_pres))$.
- Roll/pitch under **steady, level (unaccelerated) flight assumption** ($\dot{u} = \dot{v} = \dot{w} = p = q = r = 0$): the accelerometer equations reduce to $LPF(y_accel, x) = gsin\theta$, $LPF(y_accel, y) = -gcos\theta sin\varphi$, $LPF(y_accel, z) = -gcos\theta cos\varphi$, solved as: $\hat{\varphi}accel = atan^{-1}(LPF(y_accel, y)/LPF(y_accel, z))$, $\hat{\theta}accel = sin^{-1}(LPF(y_accel, x)/g)$. Demonstrated in simulation to work reasonably for p,q,r,h,Va, but to fail badly for φ,θ during maneuvers (the steady-flight assumption breaks) - motivating the EKF.
- GPS-derived states (pn, pe, χ, Vg) can also be obtained by direct LPF inversion, but GPS update rate (1 Hz) is too slow, leaving gaps to fill between updates → motivates GPS smoothing (below).

3.3 Dynamic observer theory (build-up to Kalman filter)

- LTI continuous observer: $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$ ("copy of the model" + "correction due to sensor reading"). Error dynamics $\dot{\tilde{x}} = (A - LC)\tilde{x}$ → error → 0 iff eig(A-LC) in the left half-plane.
- **Predictor-corrector** structure for sampled-data / nonlinear systems: predict between measurements ($\dot{\hat{x}} = A\hat{x} + Bu$ or $\dot{\hat{x}} = f(\hat{x}, u)$), correct at a measurement ($\hat{x}^+ = \hat{x}^- + L(y(tn) - C\hat{x}^-)$ or ... $- h(\hat{x}^-)$).

3.4 Kalman filter - full derivation (continuous-discrete form)

Model: $\dot{x} = Ax + Bu + \xi$, $y[n] = Cx[n] + \eta[n]$, $\xi\ N(0, Q)$ process noise, $\eta\ N(0, R)$ measurement noise. R usually comes from sensor calibration; **Q is the main tuning parameter**.

- **Prediction (between measurements)**: error $\dot{\tilde{x}} = A\tilde{x} + \xi \Rightarrow$ covariance ODE $\dot{P} = AP + PA^T + Q$ (derived by differentiating $E[\tilde{x}\tilde{x}^T]$ and using $E[\xi(\tau)\xi^T(t)] = Q\delta(t-\tau)$, taking half the delta-function area since integration is up to t).
- **Correction (at a measurement)**: $\tilde{x}^+ = (I-LC)\tilde{x}^- - L\eta$, $P^+ = (I-LC)P^-(I-LC)^T + LRL^T$ (Joseph form). Minimizing $tr(P^+)$ over L (using trace-derivative identities $\partial/\partial A tr(BAD) = B^T D^T$ and $\partial/\partial A tr(ABA^T) = 2AB$) gives the **optimal Kalman gain**: $L = P^-C^T(R + CP^-C^T)^{-1}$, and substituting back: $P^+ = (I - LC)P^-(I - LC)^T + LRL^T$.
- **Discretization for implementation** (Euler/2nd-order expansion): $Ad = e^(A \cdot Ts) \approx I + A \cdot Ts + A^2Ts^2/2$, giving discrete-time covariance propagation $P_{k+1} = Ad \cdot Pk \cdot Ad^T + Ts^2 \cdot Q$ (this form is guaranteed to keep P positive-definite, unlike naive first-order Euler $Pk + 1 = Pk + Ts(APk + PkA^T + Q)$ which can lose positive-definiteness numerically).
- **Full continuous-discrete KF summary** (for multiple sensors i, each processed sequentially if R is diagonal/uncorrelated):
  - Prediction: $\dot{\hat{x}} = A\hat{x} + Bu$, $Ad = I + ATs + A^2Ts^2/2$, $P_{k+1} = AdPkAd^T + Ts^2Q$.
  - Correction at sensor i: $Li = P^-Ci^T(Ri + CiP^-Ci^T)^{-1}$, $\hat{x}^+ = \hat{x}^- + Li(yi - Ci\hat{x}^-)$, $P^+ = (I - LiCi)P^-(I - LiCi)^T + LiRiLi^T$.

3.5 Extended Kalman Filter (EKF)

- For nonlinear system $\dot{x} = f(x, u) + \xi$, $yi[n] = hi(x[n], u[n]) + \eta i[n]$. Linearize about the current estimate: $A = \partial f/\partial x(\hat{x}, u)$, $Ci = \partial hi/\partial x(\hat{x}^-)$, then apply the same discretized KF recursion as above using these Jacobians in place of constant A, C.
- **Algorithm box (pseudocode, given verbatim in slides)** - "Continuous-Discrete Extended Kalman Filter":
  1. Initialize $\hat{x} = 0$.
  2. Pick output rate $Tout \ll$ sensor sample rates.
  3. At each `Tout`: run N prediction sub-steps with $Tp = Tout/N$ (this inner-loop sub-stepping improves numerical accuracy of the discretization).
  4. On receipt of a measurement from sensor i: compute `Ci`, `Li`, update `P` and $\hat{x}$ (correction step).
  - Note: if R is diagonal (sensors uncorrelated), corrections can be applied one measurement at a time sequentially rather than as a batch.

3.6 Worked example: attitude (φ, θ) estimation via EKF

- Nonlinear propagation model (from Euler-angle kinematics): $\dot{\varphi} = p + q \cdot sin\varphi \cdot tan\theta + r \cdot cos\varphi \cdot tan\theta + \xi\varphi$, $\dot{\theta} = q \cdot cos\varphi - r \cdot sin\varphi + \xi\theta$.
- Output = accelerometer, $y_a ccel = (\dot{u} + qw - rv + gsin\theta, \dot{v} + ru - pw - gcos\theta sin\varphi, \dot{w} + pv - qu - gcos\theta cos\varphi)^T + \eta_a ccel$.
- Problem: no direct measurement of $\dot{u}, \dot{v}, \dot{w}, u, v, w$. Solution: assume $\dot{u} \approx \dot{v} \approx \dot{w} \approx 0$ and approximate $(u, v, w)^T \approx Va(cos\alpha cos\beta, sin\beta, sin\alpha cos\beta)^T \approx Va(cos\theta, 0, sin\theta)^T$ (small-sideslip, α≈θ approximation) — substituting collapses the output equation to a function of (φ, θ) and the measured inputs `(p,q,r,Va)` only: $y_a ccel = (qVasin\theta + gsin\theta, rVacos\theta - pVasin\theta - gcos\theta sin\varphi, -qVacos\theta - gcos\theta cos\varphi)^T + \eta_a ccel$.
- State-space form: $x = (\varphi, \theta)^T$, $u = (p, q, r, Va)^T$, $\xi = (\xi\varphi, \xi\theta)^T$, $\eta = (\eta\varphi, \eta\theta)^T$; noise on the gyro/pressure inputs $u_a ctual = u + \xi u$ propagates through the input-noise Jacobian `G(x)`: $\dot{x} = f(x, u) + G(x)\xi u + \xi$, $y = h(x, u) + \eta$, with $G(x) = [[1, sin\varphi \cdot tan\theta, cos\varphi \cdot tan\theta, 0], [0, cos\varphi, -sin\varphi, 0]]$, and combined process noise covariance $G \cdot Qu \cdot G^T + Q$.
- Jacobians for the EKF: $A = \partial f/\partial x$ and $C = \partial h/\partial x$ given explicitly (2×2 and 3×2 matrices, trig-heavy - see slide 36 for exact entries if re-deriving).
- **Result**: EKF attitude tracking shown in simulation to be "not perfect, but significantly better" than the pure sensor-inversion approach (matches roll/pitch through maneuvers where the level-flight assumption previously broke down).

3.7 GPS smoothing

- Goal: fill in estimates between slow (1 Hz) GPS updates, and estimate wind `(wn, we)`.
- Assuming level flight, position kinematics: $\dot{p}n = Vg \cdot cos\chi$, $\dot{p}e = Vg \cdot sin\chi$.
- Ground-speed dynamics derived from the wind triangle $Vg = |Va \cdot (cos\psi, sin\psi) + (wn, we)|$, differentiated (assuming constant Va, wind) to: $\dot{V}g = [(Vacos\psi + wn)(-Va\dot{\psi}sin\psi) + (Vasin\psi + we)(Va\dot{\psi}cos\psi)]/Vg$.
- Course-angle dynamics (from a coordinated-turn assumption): $\dot{\chi} = (g/Vg) \cdot tan\varphi \cdot \cos(\chi - \psi)$.
- Wind assumed constant: $\dot{w}n = 0$, $\dot{w}e = 0$. Heading kinematics: $\dot{\psi} = q \cdot sin\varphi/cos\theta + r \cdot cos\varphi/cos\theta$.
- Full state $x = (pn, pe, Vg, \chi, wn, we, \psi)^T$, input $u = (Va, q, r, \varphi, \theta)^T$; `f(x,u)` and its Jacobian $\partial f/\partial x$ given explicitly in slides (sparse structure - wind and ψ evolve independently of position/speed states at first order).
- **Pseudo-measurements from the wind-triangle constraint**: since $(pn, pe, Vg, \chi, wn, we, \psi)$ are not independent (related by the wind triangle), two synthetic zero-valued measurements are added to make the system observable: $y_w indtri, n = Vacos\psi + wn - Vgcos\chi$ (≡0), $y_w indtri, e = Vasin\psi + we - Vgsin\chi$ (≡0).
- Combined measurement vector $yGPS = (yGPS, n, yGPS, e, yGPS, Vg, yGPS, \chi, ywind,$ with `h(x,u)` and its Jacobian given explicitly (slide 43) - a 6×7 matrix, mostly identity/trig blocks.

- **Result**: GPS-smoothed estimates of position, ground speed, course, wind, and heading track the true trajectory closely in simulation (small errors, a few meters / few m/s / few degrees).

3.8 Measurement gating (outlier rejection)

- Innovation sequence $\nu k = yk - C\hat{x}k^{-} = C\tilde{x}k^{-} + \eta k$, with covariance $Sk = R + CPk^{-}C^{T}$.
- Test statistic $zk = (yk - h(xk))^{T} \cdot Sk^{-1} \cdot (yk - h(xk))$ is **chi-squared distributed with m degrees of freedom** (m = measurement dimension). A measurement update is only accepted if `zk` falls below a chi-squared threshold at a chosen confidence level (e.g. code snippet shown uses $scipy.stats.chi2.isf(q = 0.01, df = 3)$ for a 3-dof accelerometer measurement) - this rejects statistical outliers before they corrupt the filter.

3.9 Practical tuning advice (given verbatim, useful checklist)

1. Implement the EKF in stages: attitude estimator first, then GPS smoother.
2. Test/tune each component independently and thoroughly.
3. First verify the filter tracks states correctly with **noise-free** simulated sensors (exposes coding bugs, shows the filter's best-case performance ceiling).
4. Keep wind at zero until the base filter is confidently tuned.
5. `R` is usually well known from sensor datasheets/calibration; **tune primarily via `Q`**.
6. Tune state-by-state: excite one state at a time, keep others quiet, adjust the corresponding `Q` entry by trial and error.
7. Avoid extreme/abrupt inputs while tuning (large steps) - the filter's dynamic response has physical limits; jerky truth trajectories are inherently hard to estimate.
8. Re-check your equations - repeatedly.

4 Key equations reference

- Accelerometer (spring-mass): $m \cdot \ddot{\delta} + \beta \cdot \dot{\delta} + k \cdot \delta = m(\ddot{z} - g)$; specific-force interpretation $a = (1/m)(Ftotal - Fgravity)$ - **accelerometers never read gravity directly**.
- Body-frame accelerometer output: $[ax, ay, az] = [\dot{u} + qw - rv + gsin\theta, \dot{v} + ru - pw - gcos\theta sin\varphi, \dot{w} + pv - qu - gcos\theta cos\varphi]$.
- Rate gyro (Coriolis): $aC = 2\Omega \times v$; sensor output $\propto K \cdot \Omega$ plus bias+noise $y_g yro = p(orq, r) + \beta + \eta$.
- Barometric altitude: $y_a bs_p res = \rho g \cdot h_A GL + \beta + \eta$; airspeed from Bernoulli: $y_d iff_p res = \rho Va^2/2 + \beta + \eta$.
- Magnetometer heading: $\psi = \delta + \psi m$, $\psi m = -\text{atan2}(m0y^{v}1, m0x^{v}1)$ after compensating for roll/pitch.

- GPS: 4 satellites solve for (lat,lon,alt,clock offset); error ≈ $DOP \times UERE$; error modeled as first-order Gauss-Markov process $\nu[n+1] = e^{-kTs}\nu[n] + \eta[n]$.
- LQR/Kalman duality: LQR gain $K = R^{-1}B^T P$ solves $PA + A^T P + Q - PBR^{-1}B^T P = 0$; Kalman gain $L = \Sigma C^T N^{-1}$ solves $A\Sigma + \Sigma A^T + W - \Sigma C^T N^{-1} C\Sigma = 0$.
- Continuous-discrete Kalman filter: prediction $\dot{\hat{x}} = A\hat{x} + Bu$, $Ad = I + ATs + A^2Ts^2/2$, $P_{k+1} = AdPkAd^T + Ts^2Q$; correction $Li = P^- Ci^T (Ri + CiP^- Ci^T)^{-1}$, $\hat{x}^+ = \hat{x}^- + Li(yi - Ci\hat{x}^-)$, $P^+ = (I - LiCi)P^-(I - LiCi)^T + LiRiLi^T$.
- EKF = same recursion with $A = \partial f/\partial x(\hat{x}, u)$ and $Ci = \partial hi/\partial x(\hat{x}^-)$ (local linearization at each step).
- Complementary filter: $\hat{x} = GL(s) \cdot xmL + GH(s) \cdot xmH$, $GL(s) = l/(s+l)$, $GH(s) = s/(s+l)$, $GL + GH = 1$ - provably a special (fixed-gain) case of the Kalman filter.
- Measurement gating: reject a measurement if $(y - h(x))^T S^{-1}(y - h(x))$ exceeds a chi-squared threshold for the measurement's degrees of freedom.
- Observability matters concretely: a rate-gyro-only roll estimator (state $[\varphi, bp, br]$, output `rm` only) is **not observable**; adding an accelerometer-derived roll measurement and the roll rate `pm` makes it observable. Always check `(A,C)` before trusting an EKF/KF design.

5 Open questions / unclear points

- Course lecture slide 12 has a typo in the stochastic system definition: $y = Cy + n$ should read $y = Cx + n$ (confirmed by every other occurrence of the same model in the deck and in the textbook chapter, which consistently write $y = Cx + n$).
- The "(A, B̄) Controllable?" annotation in the LQR/KF duality slide (course lecture, slide 14) is left as an open question mark in the source material itself - the course does not give a definitive general answer for when the noise-input pair needs to be controllable versus merely stabilizable; treat as a nuance to raise with the instructor if it becomes relevant to an assignment.
- The two Beard & McLain chapters are the *generic fixed-wing MAV* treatment (states $pn, pe, pd, u, v, w, \varphi, \theta, \psi, p, q, r$); the course's own AR.Drone lab platform has no GPS and adds sonar+vision instead, so the GPS-smoothing and airspeed/pitot material in Ch6/Ch8 does not directly transfer to the drone labs - it's provided as generic background for the wider "UAV" concept (likely also relevant to a fixed-wing part of the course not covered by these three files). See `lectures-22-23/notes/` for how this connects to the rigid-body/quadrotor modeling chapters, once those are summarized in sibling notes files.
- Slide "Sensors: AR.Drone" (course lecture) does not give a quantitative model for the vision-based velocity estimate (no equations, just qualitative algorithm names) - if quantitative modeling of optical flow is needed later, it isn't in this material and would need to come from the AR.Drone devkit docs or additional papers.